

Phyloseq Visualizations in Galaxy

BIOL 4810

2026-04-16

Goal: Use **phyloseq in Galaxy** to visualize your microbial community data with **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

WCSPMicrobioime

0B. Grouping variable used for comparisons:

Sample Source

0C. Other metadata variable(s) available for plotting or faceting:

Feeces vs cloacal, infection status, and sex

phyloseq workflow focus

For this module, you will generate and interpret:

Alpha diversity → Beta diversity (NMDS using Bray–Curtis) → Taxonomic bar plot

Reminder: These visualizations help you describe diversity within samples, differences among samples, and which taxa are driving those patterns.

Part 1 — Alpha diversity

1A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity is a metric describing how diverse a community is.

1B) Observed richness

What does Observed measure?

Observed measures the species richness. It is the total number of species.

1C) Shannon diversity

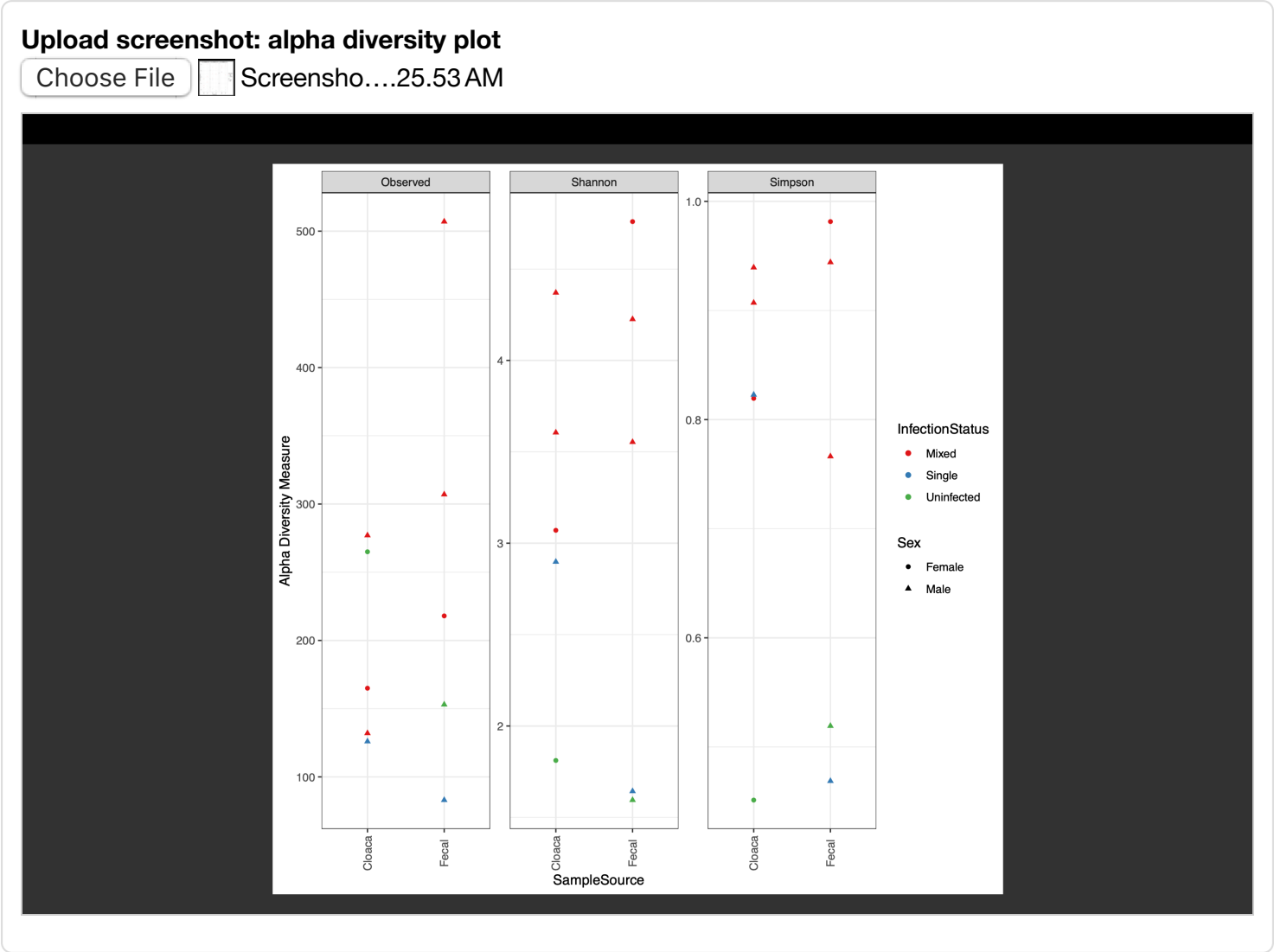
What does Shannon measure?

Shannon measures richness and evenness.

1D) Simpson diversity

What does Simpson measure?

Simpson measures richness and evenness as well. It is less sensitive to rare species, determines the probability of randomly picking the same species twice.



Grouping variable shown on the alpha diversity plot:

Sample Source

Which sample or group appears to have the greatest alpha diversity?

The fecal sample source appears to have the greatest alpha diversity.

Did Observed, Shannon, and Simpson show the same pattern?

The Observed, Shannon, and Simpson generally show the same pattern. The only difference is the scaling. Each metric measures alpha diversity differently.

Part 2 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta Diversity is the comparison of the composition of one community to another. It measures how similar or different the communities are.

2A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis measures the differences in the microbiome between two different samples.

2B) NMDS

What does NMDS show?

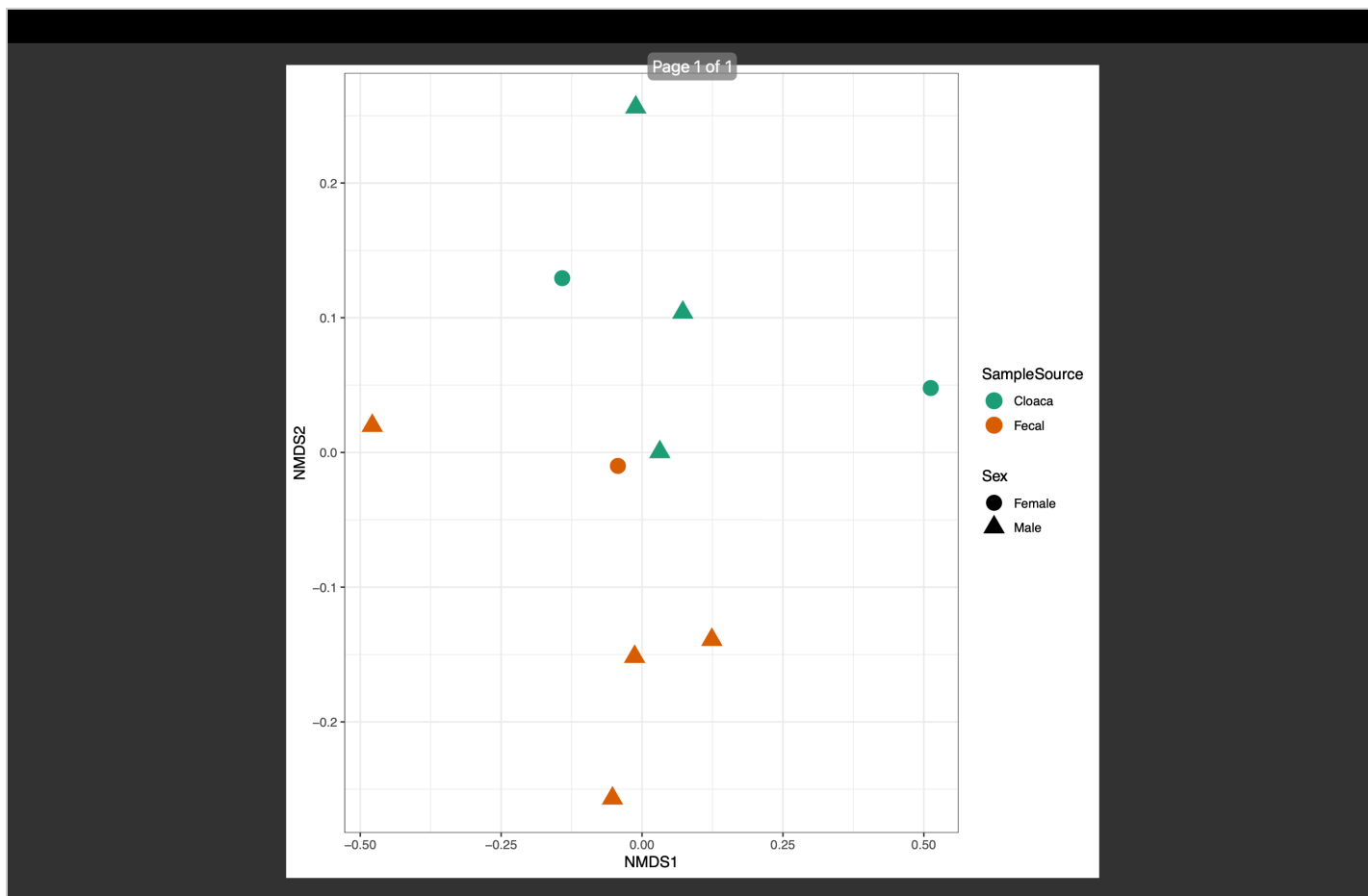
The NMDS plot shows the distance between how similar or different the microbiomes are.

Upload screenshot: NMDS Bray-Curtis plot

Choose File



Screensho....29.58 AM



Color variable used on the NMDS plot:

Sample Source and Sex

Shape variable used on the NMDS plot (if any):

Circles represent female and triangles represent males.

Which samples or groups cluster together?

The fecal samples cluster towards the bottom of the plot, while the cloaca samples cluster towards the top of the plot.

Are any groups clearly separated from one another? What might that suggest?

There are no groups that are completely separated from one another. This might suggest that the fecal and cloaca microbiomes share many similarities.

If points overlap strongly, what does that mean?

If points overlap strongly, it means that the samples have a very low Bray-Curtis dissimilarity values. It means that the samples are nearly identical.

Part 3 — Taxonomic bar plot

What a taxonomic bar plot shows:

A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Phylum

How did you facet the plot?

SampleSource

Why was this taxonomic level useful?

This taxonomic level is useful because it helps identify the difference in microbiomes without it getting too noisy. Genus and species can get overwhelming.

Why was your faceting choice useful?

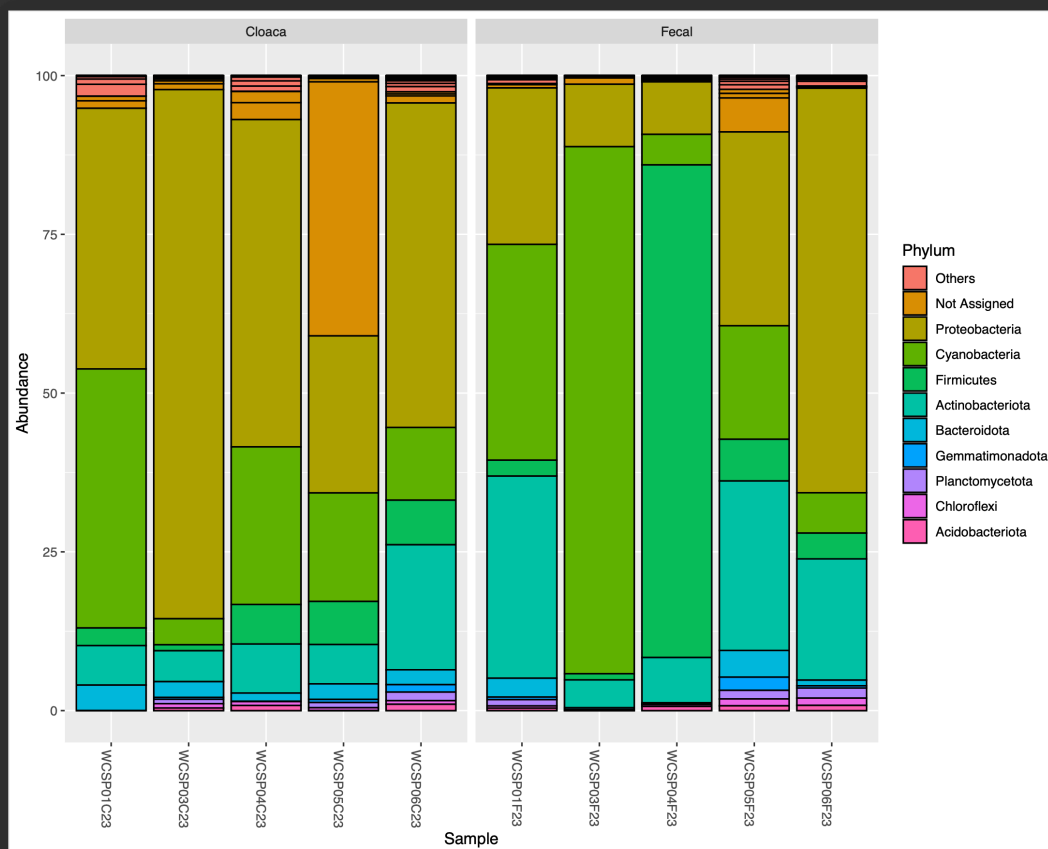
SampleSource is useful because it allows for a direct comparison between the fecal and cloaca.

Upload screenshot: taxonomic bar plot

Choose File



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Which taxa appear to be most abundant?

Proteobacteria, Firmicutes, Cyanobacteria, and, Actinobacteriota

Did taxonomic composition differ among groups?

The cloaca samples show a higher abundance of Proteobacteria and Not Assigned taxa (orange). The fecal samples tend to have more Firmicutes and Cyanobacteria, some samples (like WCSP03F23 and

Part 4 — Overall interpretation

In 3–5 sentences, summarize what your alpha diversity, beta diversity, and bar plot together suggest about this dataset.

The alpha diversity plot shows that the fecal samples are generally more "crowded" and diverse than cloaca samples. The beta diversity shows that even though fecal samples have more variety, the types of bacteria are pretty similar between both sites. The bar plot shows that both sites are dominated by Proteobacteria and Firmicutes. The fecal samples tend to have more Firmicutes, while cloaca samples have more Proteobacteria.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset

Project/sample set: WCSPMicrobioime

Grouping variable: Sample Source

Other metadata variables: Feeces vs cloacal, infection status, and sex

Alpha diversity

Definition: Alpha diversity is a metric describing how diverse a community is.

Observed: Observed measures the species richness. It is the total number of species.

Shannon: Shannon measures richness and evenness.

Simpson: Simpson measures richness and evenness as well. It is less sensitive to rare species, determines the probability of randomly picking the same species twice.

Grouping variable: Sample Source

Highest alpha diversity: The fecal sample source appears to have the greatest alpha diversity.

Did the metrics show the same pattern? The Observed, Shannon, and Simpson generally show the same pattern. The only difference is the scaling. Each metric measures alpha diversity differently.

Beta diversity (NMDS Bray-Curtis)

Definition: Beta Diversity is the comparison of the composition of one community to another. It measures how similar or different the communities are.

Bray-Curtis: Bray-Curtis measures the differences in the microbiome between two different samples.

NMDS: The NMDS plot shows the distance between how similar or different the microbiomes are.

Color variable: Sample Source and Sex

Shape variable: Circles represent female and triangles represent males.

Which groups cluster together? The fecal samples cluster towards the bottom of the plot, while the cloaca samples cluster towards the top of the plot.

Are any groups separated? There are no groups that are completely separated from one another. This might suggest that the fecal and cloaca microbiomes share many similarities.

If points overlap strongly: If points overlap strongly, it means that the samples have a very low Bray-Curtis dissimilarity values. It means that the samples are nearly identical.

Taxonomic bar plot

Taxonomic level: Phylum

Faceted by: SampleSource

Why this taxonomic level? This taxonomic level is useful because it helps identify the difference in microbiomes without it getting too noisy. Genus and species can get overwhelming.

Why this faceting choice? SampleSource is useful because it allows for a direct comparison between the fecal and cloaca.

Most abundant taxa: Proteobacteria, Firmicutes, Cyanobacteria, and, Actinobacteriota

Did composition differ among groups? The cloaca samples show a higher abundance of Proteobacteria and Not Assigned taxa (orange). The fecal samples tend to have more Firmicutes and Cyanobacteria, some samples (like WCSP03F23 and WCSP04F23) have very high concentrations of Firmicutes.

Overall interpretation

The alpha diversity plot shows that the fecal samples are generally more "crowded" and diverse than cloaca samples. The beta diversity shows that even though fecal samples have more variety, the types of bacteria are pretty similar between both sites. The bar plot shows that both sites are dominated by Proteobacteria and Firmicutes. The fecal samples tend to have more Firmicutes, while cloaca samples have more Proteobacteria.